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1.0 Introduction

Nowadays images are everywhere. We use them on our phones in systems, robotics, satellite imaging and computer graphics. Due to different screen sizes of devices and resolutions, changing the size of an image is very important in digital image processing. Changing the size of an image means making it bigger or smaller. Digital images are made up of dots that are called pixels so changing the size of an image means changing the number of pixels. When we make an image bigger, we need to guess the values of pixels to keep the image quality good and prevent deformation.

The quality of resized image depends on guessing the values of pixels. This process is called interpolation. Simple interpolation methods, like using the pixels, are fast but often make blurred or unclear images. One of these methods is called bilinear interpolation, gives results but can still make the image blurry and lose details. As a result of these problems, advanced interpolation methods were developed to improve image quality.

One of the commonly used advanced methods is **bicubic interpolation**. It differs from bilinear interpolation, instead of using four nearby pixels, bicubic interpolation uses 16 nearby pixels to get new values. It also looks at how the pixel values change, getting smoother transitions and more natural-looking results. As a result, bicubic interpolation produces smoother edges, minimize visual artifacts and keeps image details more accurately.

Keeping image quality good during resizing is still a challenging task especially in applications where fine details are important. For example, doctors often make MRI (Magnetic Resonance Imaging) or CT (Computed Tomography) scan images bigger to examine small structures clearly. Similarly in robotics and computer vision systems, accurate image scaling improves object detection and image analysis. Poor scaling techniques can reduce clarity and negatively affect system performance.

Bicubic interpolation is widely used in image editing software, computer graphics and many engineering applications because it provides a balance between image quality and computational efficiency.

In this report, we will talk about image scaling and explain the role of bicubic interpolation in improving resized images. In addition to discussing the mathematical idea behind the method and highlights its importance in modern digital imaging systems.

2.0 Fundamentals of Digital Imaging

Digital images are the basis of modern systems. These systems include imaging and remote sensing. They also include computer vision and multimedia applications. Therefore, digital images are important for these systems. The systems need to process the images efficiently. So, it is essential to understand how digital images are made. Digital images are not just pictures. Instead, they are collections of data.

The way this data is organized affects the quality of the image. It also affects how well the image can be processed. For this reason, we need to look at the concepts of digital images. These concepts include pixel representation and spatial organization. They also include resolution and sampling.

We need to understand these concepts before we can learn advanced methods such as interpolation. Digital images are the foundation of modern engineering and technological systems ranging from medical imaging and remote sensing to computer vision and multimedia applications.

2.1 Digital Image Representation

A digital image is a version of a continuous visual scene. Real-world images are analog. So, they need to be converted into a format. This conversion is done by dividing the image into units called pixels. Each pixel represents a sample of the image at a location.

It is assigned a value that describes its visual properties. For example, in grayscale images this value represents intensity or brightness. The value is usually on an 8-bit scale, which's from 0 to 255. However, color images are different. They are represented using components. These components correspond to color channels. The accuracy of this representation affects the overall image quality. It also affects how well the image can be processed. Digital images are represented in this way because real-world images exist in analog form.

2.2 Pixel Grids and Coordinate Systems

A digital image is organized as a two- grid. The grid consists of pixels arranged in rows and columns. From a perspective this structure is represented as a matrix. Each element in the matrix corresponds to a pixel value. Each pixel in the grid has a position. This position is defined using coordinate notation, which's (x, y). The coordinate notation specifies the location of the pixel along the vertical axes. The spatial framework allows image data to be accessed systematically. In most systems, the origin is at the top-left corner. The x-axis extends across columns, and the y-axis extends downward across rows. This arrangement is essential for image processing. For example, many techniques rely on the values of a pixel's neighbors. Bicubic interpolation is one of these techniques, it considers a 4 x 4 neighborhood of pixels to generate results.

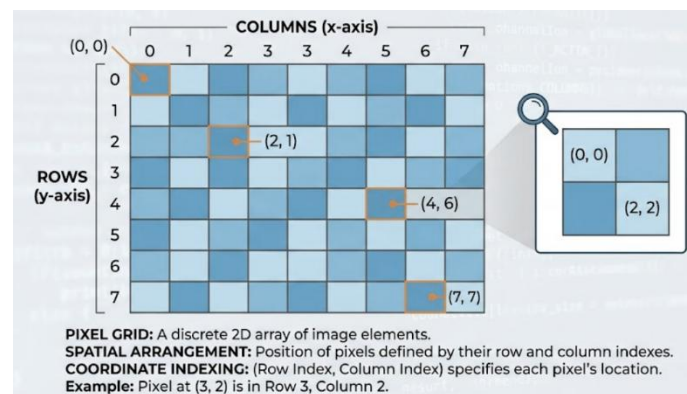


Figure 1: Spatial Arrangement and Coordinate Indexing of a 2D Pixel Grid

2.3 Resolution and Color Representation

The quality and interpretability of a digital image are strongly influenced by its resolution and the method used for color representation. These factors determine how much detail the image contains and how accurately it reflects the original scene.

Resolution:

This refers to the number of pixels used to represent an image, commonly expressed as width x height (e.g., 1920 x 1080). Higher resolution reduces the quantization error of the original scene. This is especially important in image scaling, where interpolation techniques are required to estimate new pixel values while preserving visual quality.

Color Representation:

Color in digital images is most commonly represented using the RGB color model, as shown in Figure 2. This model is based on the combination of three primary colors: Red, Green, and Blue. By varying the intensity of each component, a wide spectrum of colors can be produced.

Alternatively, images may be represented in grayscale, where each pixel is assigned a single intensity value. While grayscale images contain less information, they are often sufficient for structural analysis and significantly reduce computational complexity.

This visualization demonstrates the superposition of Red, Green, and Blue primary spectral components. In engineering applications, these are treated as three independent data channels that define the final pixel intensity and hue.

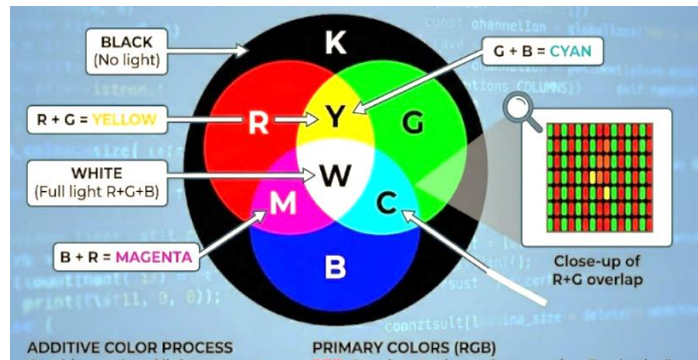


Figure 2

2.4 Sampling and the Link to Interpolation

Sampling is the process of converting a visual scene into a discrete digital representation. It is done by measuring the scene at spatial intervals. The Nyquist-Shannon Sampling Theorem states that the sampling frequency must be high enough to capture the details of the original scene. If the sampling frequency is not high enough visual distortions such as aliasing may occur. Sampling is linked to interpolation. When an image is resized or transformed the original discrete grid does not contain values for the required pixel locations. Interpolation techniques are used to estimate these values based on existing data. Bicubic interpolation produces transitions and more visually accurate results than simpler methods. It is the industry standard for high-quality image scaling. Digital images are then interpolated to produce the final image.

3.0 Interpolation

In digital signal processing, interpolation is the mathematical algorithm that allows us to move between different spatial resolutions.

3.1 What is Interpolation

Interpolation is a method of constructing new data points within the range of a discrete set of known data points. In digital imaging, it is the process of calculating the properties (color and intensity) of a pixel that does not exist in the source but is required for the new, transformed version of the image.

Mathematically, if we have a set of points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, interpolation finds a function $f(x)$ such that $f(x_i) = y_i$ for all i . For any value of x between $x_0 \leq x \leq x_n$, the function provides a calculated estimate that follows the trend of the surrounding data.

3.2 Why Is It Needed

Interpolation becomes a necessity the moment we move away from an image's "native resolution." It is required in the following scenarios:

- **Image Upscaling:** When enlarging a $100 \times 100 \rightarrow 500 \times 500$, the computer must create 240,000 new pixels. Interpolation decides what color those pixels should be so the image doesn't look like a collection of disjointed squares.
- **Geometric Transformations:** When an image is rotated (e.g., by 15°), the original square pixels are tilted. The new grid points no longer align with the original ones, requiring a calculation to redistribute the color values.
- **Sub-Pixel Registration:** In medical imaging and satellite tracking, features often move by fractions of a pixel. Interpolation allows the system to track movement at a higher precision than the hardware sensor originally captured.

3.3 Estimating/Correcting Unknown Pixel Values

The primary goal of interpolation is to maintain the visual integrity of an image during transformation.

- **Estimation:** By using a mathematical function (the kernel), the system calculates a weighted average of neighbor pixels to "guess" the value of a new pixel.
- **Correction:** It can be used to restore missing data in corrupted images or to smooth out "staircase" effects (aliasing) that occur when digital signals are poorly sampled.

3.4 Continuous vs. Discrete System Data

To understand how bicubic interpolation works, one must understand the shift between two types of data:

Discrete Data (The Digital Reality): Digital images are discrete. They are a matrix of "samples" taken at specific intervals. Between Pixel $(0,0)$, $(1,0)$, There is no data; it is a void.

Continuous Data (The Physical Reality): The light hitting a camera sensor is a continuous wave.

The Interpolation Goal: The algorithm treats the discrete pixels as if they were samples from a continuous mathematical surface. By fitting a "surface" or "curve" over the discrete points, we can sample that surface at any coordinate—even at $(0.5, 0.5)$ to generate a new pixel.

Interpolation essentially "re-maps" discrete points back into a continuous function.

3.5 1D vs. 2D Interpolation

Feature	1D Interpolation	2D Interpolation
Data Structure	Operates on a linear array or sequence of data points.	Operates on a grid or matrix (spatial coordinates).
Dimensionality	Single axis x .	Dual axes (x, y) .
Typical Data	Audio signals, sensor logs, or single-line graphs.	Digital images, topographical maps, and video frames.
Neighbors Used	Uses immediate neighbors to the left and right (e.g., 2 points for linear, 4 for cubic).	Uses a surrounding "neighborhood" of pixels (e.g., 2×2 for bilinear, 4×4 for bicubic).
Core Calculation	Calculates the value of y for a given x .	Calculates the intensity z (or color) for a given (x, y) .
Visual Result	Results in a smooth curve or line transition.	Results in a smooth surface or image transition.
Complexity	Low; involves solving a single polynomial equation.	High; often requires "separable" passes (interpolating rows then columns).
Interpolation Type	Linear, Cubic Spline, Polynomial.	Bilinear, Bicubic, Lanczos, Thin-plate Spline.

4.0 Types of Interpolation Methods

Image Interpolation is a fundamental process in digital imaging used to estimate unknown pixel values when an image is transformed. Since images are composed of discrete pixels, scaling an image requires mathematical methods to generate new pixel values while preserving image quality.

4.1 Nearest Neighbor Interpolation

Nearest neighbor interpolation is the simplest image scaling technique. It assigns the value of the closest original pixel to each newly created pixel without using any averaging or smoothing as opposed to other methods.

How it works:

When enlarging an image, each new pixel copies the value of the nearest source pixel.

Advantages:

- Very fast computing.
- Easy to implement.

- Low memory usage.

Disadvantages:

- Produces jagged edges.
- Poor image quality when enlarged.
- Loss of smooth details.

Applications:

- Pixel art scaling.
- Simple embedded systems.
- Systems requiring speed.

4.2 Bilinear Interpolation

Bilinear interpolation improves image quality by using the weighed average of the four nearest surrounding pixels.

How it works:

It performs linear interpolation both horizontally and vertically to estimate the new pixel value.

Advantages:

- Smoother pixel quality compared to 'nearest neighbor'.
- Moderate computational cost.
- Widely supported

Disadvantages:

- Can blur edges.
- Reduced sharpness compared to cubic.

Applications:

- Basic image viewers.
- Web image scaling.
- Real-time graphics.

4.3 Bicubic Interpolation

Bicubic interpolation is a more advanced technique that uses the nearest 16 surrounding pixels arranged in a 4x4 pixel grid to calculate the new pixel.

How it works:

It uses the cubic interpolation method on the 2 directions (x and y axes) in a 4x4 square to calculate new pixel values.

Advantages:

- High image quality.
- Smooth gradients and less banding.

- Better detail retention than bilinear.

Disadvantages:

- Relatively slow compared to bilinear and nearest neighbor.
- Uses more computer resources.

Applications:

- Professional photography and photo editing.
- Medical imaging.
- Satellite imagery.

4.4 Lanczos Method

Lanczos interpolation is a high-quality resampling method based on sinc functions ($\frac{\sin x}{x}$).

How it works:

It uses larger neighborhoods of surrounding pixels and advanced weighing techniques to preserve details.

Advantages:

- Excellent sharpness.
- Strong edge preservation.

Disadvantages:

- Extremely slow and computationally expensive.
- Can create ringing artifacts.
- More mathematically complex.

Applications:

- Professional image processing
- Video scaling

5.0 Bicubic Interpolation Theory

5.1 Definition of Bicubic Interpolation

Bicubic interpolation is an image resampling technique that estimates the intensity of a new pixel using a weighted combination of the closest 16 neighboring pixels. It is based on cubic polynomial interpolation applied in two dimensions. Albeit being more advanced, it provides smoother and more visually appealing results than generic interpolation.

5.2 Cubic Interpolation in 1D vs 2D

1D Cubic Interpolation

In one-dimension, cubic interpolation estimates unknown values on a line using four nearby data points. Example: $ax^3 + bx^2 + cx + d$. It is used when values change along one axis only.

2D Bicubic Interpolation

This is the more commonly used since images have a width and height, therefore, interpolation must occur in the two dimensions simultaneously. It uses the bicubic interpolation across the x dimension and the y dimension, which creates a smooth surface rather than a simple line.

5.3 Why 4x4 Pixel Grids Are Used

A cubic function requires multiple surrounding points to estimate smooth curvatures accurately in the 2-dimensional plane (4x4 grid). This preserves gradients better as well as produces smoother edges compared to methods using fewer pixels.

5.4 Smoothness and Continuity

One major advantage of bicubic interpolation is smooth transitions between pixels, this results in gradual pixel intensity changes rather than banding and no sudden jumps between neighboring pixels. This reduces jagged edges, pixelation, and harsh color transitions. As a result, enlarged images appear more natural and less pixelated.

5.5 comparison to other methods

The more pixels a method uses, the more computer resources it uses and as a result, the more time needed to process data, however, it results in better quality and less banding.

Nearest neighbor comes off as the easiest to process and produces the least quality, while Lanczos method is the hardest to process and produces the sharpest results, with bilinear and bicubic come off as 2nd and 3rd place respectively.

Bicubic interpolation offers an excellent balance between processing speed and image quality, which is why it is one of the most popular methods in digital imaging.

6.0 Mathematical Formulation of Bicubic Interpolation

6.1 Cubic Polynomial Equation

Cubic interpolation uses a cubic polynomial to estimate values between two points.

The general form of 3rd degree polynomial in one dimension is : $f(x) = ax^3 + bx^2 + cx + d$

The coefficients are calculated by determining 4 surrounding data points (two near the target and two much further away). This enables the technique to provide smoother transitions than linear interpolation and capture the slope of the data.

6.2 Bicubic surface equations

Bicubic Interpolation is an expansion of the cubic interpolation into two dimensions, it creates a continuous smooth surface with continuous derivatives. Bicubic interpolation shows the image intensity in a small 4×4 group of pixels using a cubic polynomial with two variables (x,y), The interpolated value is commonly written as:

$$f(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

Expanding this expression gives a 16-term polynomial:

$$\begin{aligned} p(x, y) = & a_{00} + a_{10}x + a_{01}y + a_{20}x^2 + a_{11}xy + a_{02}y^2 \\ & + a_{21}x^2y + a_{12}xy^2 + a_{22}x^2y^2 + a_{30}x^3 + a_{03}y^3 \\ & + a_{31}x^3y + a_{13}xy^3 + a_{32}x^3y^2 + a_{23}x^2y^3 + a_{33}x^3y^3 \end{aligned}$$

This equation contains 16 coefficients (a_{ij}). These coefficients are calculated by using the four corner pixels and the changes in their values, the partial derivative in the x and y directions and the cross derivatives (twists) at the 4 corners.

6.3 Weight coefficients

Bicubic Interpolation does not treat all 16 pixels equally, each pixel is assigned a weight according to its closeness to the target location, pixels that are closer have greater effect than those that are farther away. The distance in vertical as well as horizontal directions affects these weights. This is calculated using a simple formula in which the 2D weight is the product of two 1D weights:

$$W(x, y) = w(x) \cdot w(y)$$

This helps make the calculations easier and produces a smoother result.

6.4 Kernel Functions

In bicubic interpolation, the weights are defined by a kernel function, which is often a smooth cubic function. This kernel helps the image stay smooth without abrupt changes by giving weights to close pixels and disregarding pixels that are further away. A widely used kernel is the one introduced by Robert G. Keys, defined as:

$$W(x) = \begin{cases} (a+2)|x|^3 - (a+3)|x|^2 + 1, & |x| \leq 1 \\ a|x|^3 - 5a|x|^2 + 8a|x| - 4a, & 1 < |x| < 2 \\ 0, & \text{otherwise} \end{cases}$$

6.5 Partial Derivatives and Continuity Conditions

The concept of partial derivatives indicates the rate of variation in image intensity in various directions. For instance, the partial derivative in the x direction indicates variation horizontally, while that in the y direction indicates variation vertically. The cross-derivative, on the other hand, refers to the interaction of these two directional variations. These parameters are normally calculated using nearby pixel information and help in improving the accuracy of interpolation, especially in maintaining edges and smooth regions.

One of the strengths of bicubic interpolation is the smoothness property, also called the C^1 continuity. This indicates that across pixel boundaries, both the pixel values and their first derivatives stay continuous. Using the intensity values: $f(x, y)$, gradients: $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ and cross-derivatives: $\frac{\partial^2 f}{\partial x \partial y}$. The technique produces a smooth transition without any abrupt change in the output image. The output will have less blocky effects and appear much more natural than other interpolation techniques.

8.0 Implementation Using Tools

In this section, bicubic interpolation is implemented using two widely used tools: MATLAB and Python. Both platforms have built-in functions that simplify the interpolation process while applying the mathematical formulas of interpolation. In addition, a comparison is performed using nearest neighbor, bilinear, and bicubic methods to show the differences in image quality.

For simplicity and ease of access purposes, we included both the MATLAB code and the Python codes on a GitHub repository, the link for each code will be provided under each section.

8.1 MATLAB Implementation

[MATLAB Code](#)

The MATLAB code we wrote reads an image, enlarges it by a factor of 2 using three interpolation methods (nearest neighbor, bilinear, and bicubic), displays the results, and saves the output images.

The example for this code in action can be seen in this image:

- [MATLAB implementation Example](#)

Explanation:

The 'imresize' function in MATLAB applies different interpolation techniques depending on the selected option. The 'bicubic' method uses cubic convolution based on a 4×4 neighborhood of pixels, producing smoother and higher-quality results compared to simpler methods.

8.2 Python Implementation

[Python Code](#)

The Python code performs image resizing using OpenCV and compares the three interpolation methods.

The single image was resized using the 4 different methods and the difference can be seen in these 2 images:

- [Zoomed-in picture of the Python Implementation](#)
- [Zoomed-out picture of the Python Implementation](#)

Comparison Note:

The same image was resized using three interpolation methods to allow for visual comparison. As discussed earlier in sections 4.0 and 5.0, nearest neighbor produces a blocky and pixelated result, bilinear interpolation improves smoothness, while bicubic interpolation gives the best balance between sharpness and smooth transitions.

9.0 Image Quality Evaluation Methods

Understanding the performance of interpolation algorithms, specifically Bicubic Interpolation, requires moving beyond simple visual inspection toward quantitative analysis. While Bicubic interpolation offers a smoother result than nearest-neighbor or bilinear methods by considering a 4×4 neighborhood of pixels and using cubic splines, it can still introduce artifacts like blurring or "ringing" around sharp edges. To objectively measure how well an interpolated image preserves the integrity of the original, researchers utilize Image Quality Assessment (IQA) metrics. These metrics provide a mathematical framework to evaluate fidelity, noise, and structural similarity, allowing us to determine if the 16-pixel weighted average used in Bicubic interpolation actually maintains the perceptual essence of the high-resolution source. Bicubic interpolation cannot recover lost high-frequency details; it only estimates them based on surrounding pixel information. The choice of evaluation metric ultimately depends on whether the goal is numerical accuracy or perceptual realism.

9.1 Mean-Squared-Error (MSE)

The Mean Squared Error is the most fundamental estimator used in signal processing to quantify the difference between an original image (I) and its interpolated or reconstructed version (K). It measures the average of the squares of the "errors," that is, the pixel-by-pixel difference between the two images.

For two images of size $m \times n$, the formula is defined as:

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i,j) - K(i,j)]^2$$

Sensitivity:

MSE is highly sensitive to outliers because it squares the error. If the Bicubic algorithm significantly miscalculates a sharp edge (causing a large pixel value difference), the MSE will rise sharply.

Smoothing vs. Ringing:

The method's 16-pixel neighborhood tends to smooth transitions, lowering large pixel errors compared to nearest-neighbor. However, at sharp edges, Bicubic can "overshoot", producing ringing artifacts that disproportionately increase the MSE.

Limitations:

While mathematically convenient for optimization, MSE doesn't always correlate with human vision. A "blurry" Bicubic image might have the same MSE as a "noisy" image, even though a human would prefer the former. A low MSE does not guarantee perceptual quality.

9.2 Peak Signal-to-Noise Ratio (PSNR)

The Peak Signal-to-Noise Ratio is an expression for the ratio between the maximum possible power of a signal and the power of corrupting noise that affects the fidelity of its representation. It is derived directly from the MSE and is usually expressed in terms of the logarithmic decibel (dB) scale. The formula for PSNR is:

$$10 \times \log_{10} \frac{MAX_I^2}{MSE}$$

- Where Max_I is the maximum pixel value of the image.

Interpretation:

A higher PSNR generally indicates a higher quality reconstruction. In evaluating Bicubic interpolation, PSNR values typically fall between 30 and 50 dB.

The Logarithmic Benefit:

Because the human ear and eye perceive intensity logarithmically, the PSNR is often more intuitive than raw MSE. Small improvements in MSE often translate to a more readable increase in PSNR.

Limitations:

Like MSE, it focuses on pixel-level fidelity rather than the "meaning" of the shapes within the image. It doesn't distinguish how the error appears, only how large it is, meaning a blurry Bicubic result and a sharp but noisy result could share the same PSNR.

9.3 Structural Similarity Index (SSIM)

The Structural Similarity Index is a more sophisticated, perception-based model that departs from the error-summation philosophy of MSE and PSNR. SSIM considers image degradation as perceived change in structural information, under the assumption that the human visual system (HVS) is highly adapted for extracting structural information from a scene. SSIM compares two images based on three distinct components: Luminance (l), Contrast (c), and Structure (s).

The formula for the index between images x and y is:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Where:

- μ_x, μ_y : The pixel sample means (luminance)
- σ_x^2, σ_y^2 : The variances of x and y (contrast)
- σ_{xy} : The covariance of x and y (structural correlation)
- C_1, C_2 : Constants used to stabilize the division when the denominator is near zero.

Bicubic interpolation often results in a "smooth" look that preserves the general structure of the image better than simpler methods. SSIM captures this by looking at how local patterns of pixel intensities are preserved. Even if edges are slightly blurred, the structural relationships remain intact, which SSIM captures more effectively than MSE/PSNR. The resulting value is a decimal between -1 and 1:

- **SSIM ≈ 1 :** Nearly identical structure.
- **SSIM > 0.9 :** Very high similarity.
- **SSIM < 0.7 :** Noticeable degradation.

SSIM is often the "gold standard" for reporting results because it aligns much more closely with how a person would actually rank image quality. For color images, these metrics are typically computed for each channel and then averaged to produce a final value.

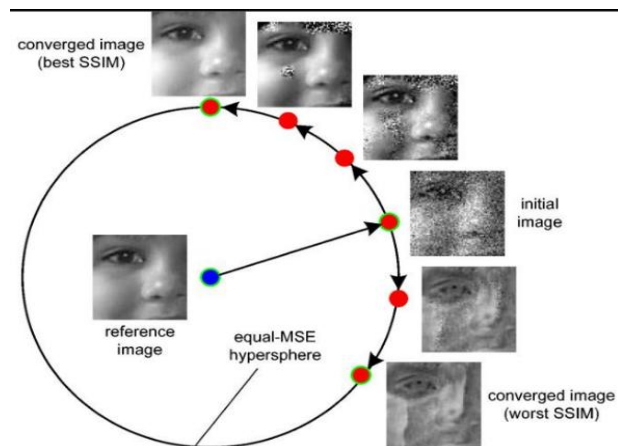


Figure 3

10.0 Applications of Bicubic Interpolation

10.1 Digital Image Processing

The bicubic interpolation algorithm addressed is a simplified computation complexity version of an algorithm. The proposed architecture is real-time and applicable in many security and surveillance applications, such as identifying the car plates, authentications, remote sensing, and similar practices, where real-time processing is necessary. The bicubic interpolation method efforts to fit a surface among four corner pixels using a third-order polynomial function. In order to compute a bicubic interpolation, the intensity values and the horizontal, vertical, and diagonal derivate at the four corner points should be calculated.

10.2 Medical Imaging

Image interpolation is the most basic requirement for many image processing tasks such as medical image processing. Image interpolation is a technique used in resizing an image. To change the image size, each pixel in the new image must be remapped to a location in the old image to calculate the new pixel value. There are many algorithms available for determining the new pixel value, most of which involve some form of interpolation between the closest pixels in the old image. Bicubic interpolation algorithm can be used to change the size of medical images from the Messidor dataset and then analyze it by measuring it using three parameters Mean Square Error (MSE), Root Mean Squared Error (RMSE), and Peak Signal-to Noise Ratio (PSNR). The Bicubic interpolation method considers the known 4x4 neighboring pixels totaling 16 pixels. Closer pixels are given a higher weight in calculations and are the ideal combination of processing time and output quality. For this reason, this method is standard in many image editing programs including Adobe Photoshop, printer drivers, and in-camera interpolation.

10.3 Geographic Information System

In recent decades, a number of regional and global digital elevation models (DEMs) have been used by researchers who need to choose between several of these models to perform their studies and to use these DEMs as third-party data to compute derived products. The bicubic resampling method is the two-dimensional extension of the cubic interpolation. It considers a 4×4 window of pixels around the coordinates of interpolation. The values of these pixels are weighted according to their position (dx, dy) relative to the input point. The impact of the bicubic parameter can be clearly observed on DEM slope discontinuities, such as mountain ridges or land-use changes. In such areas, the lowest values of the bicubic parameter increase the maximum of elevation, whereas the highest parameters do the opposite. Moreover, the maximum elevation is retrieved at different curvilinear distances depending on the bicubic parameter.

11.0 Bicubic Interpolation Assessment

In the leading sections, the meaning, applications, and mathematical fundamentals of Bicubic interpolation have been discussed. However, there still needs to be some kind of assessment to assess the quality and efficiency of bicubic interpolation.

11.1 Advantages

Bicubic interpolation works by using a $4 * 4$ grid of 16 surrounding pixels to calculate the value of a new pixel using cubic polynomials.

Smooth Gradients: By considering a larger neighborhood than simpler methods, it produces significantly smoother transitions, making it ideal for photos with soft focus or natural landscapes.

Edge Acutance: It naturally creates a slight "overshoot" effect, which increases the perceived sharpness (acutance) of edges without requiring a separate sharpening filter.

Predictability: Because it is a fixed mathematical formula, the output is deterministic and objective. It does not "guess" or add information that wasn't there, which is critical for medical imaging and scientific data.

Computational Efficiency: It is extremely fast on modern hardware. Even high-resolution 4K images can be bicubically upscaled in milliseconds on a standard CPU.

11.2 Challenges and Limitations

Ringling Artifacts (Halos): The "overshoot" that creates sharpness can also cause "ringing" faint ghost-like lines or halos around high-contrast edges (e.g., dark text on a white background).

Blurring Fine Detail: While it smooths gradients well, it cannot "invent" high-frequency details. At high magnification levels (e.g., 4x or 8x), the image begins to look "plastic" or blurry.

Aliasing: On very fine, repetitive patterns (like a brick wall or a plaid shirt), bicubic interpolation can produce Moiré patterns or jagged edges.

11.3 Bicubic vs. AI

The fundamental difference is that **Bicubic interpolation is "blind"** (it only sees math), while **AI is "informed"** (it has seen millions of images and knows what a leaf or a face *should* look like).

11.3.1 Cost Vs. Quality

Feature	Bicubic Interpolation	AI (Super-Resolution)
Visual Quality	Moderate; smooth but can be soft.	Superior; creates sharp, realistic textures.
Computational Cost	Very Low; runs on any device.	High; usually requires a GPU (NPU).
Fidelity	High; mathematically accurate source.	Variable; may "hallucinate" fake details.
Implementation	Easy; built into almost all libraries.	Complex; requires model hosting/loading.

11.3.2 ESRGAN / Super Resolution AI

ESRGAN (Enhanced Super-Resolution Generative Adversarial Network) is a leading AI model for upscaling.

How it differs: While Bicubic interpolates between existing points, ESRGAN uses a "Discriminator" network that has been trained to distinguish between real high-resolution images and fakes.

The Result: It doesn't just smooth edges; it reconstructs textures, adding individual hairs to a portrait or grain to a wooden table that were physically lost in the low-resolution original.

11.3.3 Processing Power

Bicubic: Can be processed in real-time for video streams on a smartphone or even a low-power smart TV chip. It uses standard arithmetic operations.

AI (ESRGAN): Requires massive parallel processing. While modern AI-specific hardware (like NVIDIA's Tensor cores or Apple's Neural Engine) has made this faster, it still consumes significantly more electricity and generates more heat than traditional interpolation. Upscaling a single image with AI can take 100x to 1000x longer than doing it bicubically.

12.0 Future Improvements

As display resolutions push toward 8K and beyond, the demand for high-fidelity upscaling has outpaced the capabilities of static algorithms. Future improvements in interpolation aim to bridge the gap between traditional mathematical rigor and the dynamic requirements of modern hardware. By moving away from "one-size-fits-all" kernels, researchers are focusing on architectures that can recognize local image context, utilize parallel hardware architectures, and

leverage the predictive power of neural networks to estimate high-frequency details that were previously lost. These advancements are particularly important in applications such as real-time video streaming, gaming, and medical imaging, where both accuracy and speed are critical. For example, modern video streaming platforms dynamically switch between interpolation methods depending on bandwidth and device capability. Future methods are increasingly evaluated using perceptual metrics such as SSIM rather than purely numerical metrics like MSE or PSNR, reflecting a shift toward human-centered image quality assessment. Future research may also focus on fully learned interpolation models that eliminate fixed kernels entirely, allowing systems to adapt purely through data-driven optimization.

12.1 GPU Acceleration & Texture Filtering

While Bicubic interpolation is more computationally intensive than Bilinear, modern GPU acceleration has largely mitigated this bottleneck.

Hardware-Level Implementation: Modern Graphics Processing Units (GPUs) use dedicated hardware units for texture filtering. By rewriting the Bicubic weight equations into a sum of weighted linear interpolations, the process can be offloaded to fast bilinear texture samplers.

Parallel Execution: Unlike CPU-bound processes that calculate pixel values sequentially, GPUs process thousands of 4×4 neighborhoods simultaneously. This allows for high-quality Bicubic upscaling in 2026-era gaming and video streaming at 144Hz with negligible latency.

This is why Bicubic interpolation is still widely used in graphics APIs, where performance and visual quality must be balanced in real time.

12.2 Adaptive Interpolation

The primary weakness of standard Bicubic interpolation is its uniformity, it applies the same 16pixel weight regardless of whether it is processing a flat sky or a detailed texture. Adaptive methods solve this by:

Edge-Directed Interpolation (EDI): These algorithms analyze the local gradient before interpolating. If an edge is detected, the algorithm adjusts the interpolation kernel to follow the direction of the edge rather than crossing it, significantly reducing the "ringing" and "blurring" artifacts discussed earlier.

Content-Aware Kernels: Future versions of Bicubic kernels can dynamically adjust their b and c parameters (the "stiffness" of the spline) based on local contrast, preserving sharpness in high detail areas while maintaining smoothness in gradients.

Even though adaptive methods are more accurate, they introduce additional computational overhead and decision logic, making them harder to optimize for real-time systems.

12.3 Hybrid AI + Bicubic Methods

In the current landscape of 2026, the industry is moving away from "AI-only" upscaling (which is slow and power-hungry) toward Hybrid Architectures.

The "Guided" Upscale: In these models, a fast Bicubic pass provides the "base" structural upscaling. A lightweight Neural Network (CNN or Transformer) then acts as a residual layer,

only calculating the "difference" needed to restore high-frequency textures like hair, fabric, or text.

Benefits: This hybrid approach maintains the mathematical stability and speed of Bicubic interpolation while achieving the visual "crispness" of Deep Learning Super-Resolution (DLSR), resulting in high SSIM scores without the high computational cost.

While powerful, these methods can sometimes introduce artificial details that were not present in the original image, raising questions about authenticity in critical fields like medical or forensic imaging.

12.4 Real-time Mobile Optimization

For mobile devices, power efficiency is as important as visual quality. Modern mobile optimization focuses on:

NPU Integration: Moving interpolation tasks from the general-purpose CPU to dedicated Neural Processing Units (NPUs) or specialized DSPs (Digital Signal Processors).

Variable Rate Upscaling: To save battery, mobile OSs can now apply high-quality Bicubic interpolation only to the areas of the screen where the user is focusing (foveated rendering) or specifically for media playback, while using simpler methods for static UI elements.

Real-World Constraint: Thermal limits and battery capacity remain the biggest barriers to deploying high-quality interpolation continuously on mobile platforms.

13.0 Conclusion

This report presented bicubic interpolation – a method used to enlarge or shrink images. Bicubic interpolation works by looking at 16 surrounding pixels (a 4×4 grid) to estimate the value of a new pixel. This produces smoother and sharper images than simpler methods like nearest neighbor or bilinear interpolation.

The method has several advantages. It uses cubic polynomials to create smooth transitions between pixels, and it is easy to use because programs like MATLAB and Python have built-in functions for it. The results look much better than basic interpolation methods.

However, bicubic interpolation also has limitations. When you zoom in too much (more than 4×), it cannot recreate missing details. It also requires more computing power than nearest neighbor or bilinear methods.

Compared to AI methods like ESRGAN, bicubic interpolation is much faster and simpler. AI can create very detailed images, but it is slow and needs powerful computers. Bicubic interpolation is better for real-time applications.

In summary, Bicubic interpolation is a reliable and practical way to resize images. It is a good balance between image quality and computational efficiency and resources needed. It is widely used in photography, medical imaging, and satellite imaging.

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